

Velaga Venkata Karthik

Data Science | Machine Learning | Python | Deep Learning & NLP

Vadodara, Gujarat, India | +91 9666361165 | velagavenkatakarthik@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

B.Tech Computer Science (AI) student at Parul University with practical project experience in Python, Machine Learning, Deep Learning, and NLP. Built ML and NLP workflows involving data preprocessing, feature engineering, model training, fine-tuning, and evaluation using Scikit-learn, TensorFlow, PyTorch, and Hugging Face Transformers. Seeking a Data Science or Machine Learning internship.

EDUCATION

B.Tech in Computer Science & Engineering (Artificial Intelligence)

Parul University, Vadodara, Gujarat | CGPA: 7.06/10 | Expected Graduation: 2027

TECHNICAL SKILLS

Programming Languages: Python, Java (DSA), SQL

Data Science: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, EDA, Data Cleaning, Data Preprocessing, Feature Engineering, Model Evaluation

Machine Learning: Regression, Classification, Clustering, Decision Trees, Random Forest, SVM, KNN, K-Means, PCA, Model Evaluation, Cross-Validation, Hyperparameter Tuning

Deep Learning & NLP: TensorFlow, Keras, PyTorch, Neural Networks, CNN, RNN, LSTM, GRU, NLTK, spaCy, Text Preprocessing, Word Embeddings, Transformers, T5, Sentiment Analysis, Text Classification

Generative AI: Gemini AI, Prompt Engineering, LLMs, RAG

Web & Cloud: FastAPI, React, Node.js, Express.js, AZ-900

Databases: MySQL, MongoDB

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Linux

PROJECTS

AI-Based Event Management Platform — Full Stack + Generative AI

github.com/Karthikvenkat1/AI-Based-Event-Management-Platform

Tech Stack: React.js, Next.js, Node.js, Express.js, MongoDB, Gemini AI

- Built a full-stack event management platform supporting event discovery, registration, and QR-based ticket booking.
- Integrated Gemini AI to power smart event creation and personalized recommendations.
- Implemented QR ticket verification to prevent fake entries and developed REST APIs with an organizer dashboard for managing events and bookings.

Text Summarization Using T5 — NLP | Deep Learning

github.com/Karthikvenkat1/Text-Summarization-Using-T5

Tech Stack: Python, PyTorch, Hugging Face Transformers, T5, Pandas, NLP

- Developed an abstractive text summarization system using the T5-small Transformer model on the SAMSum dialogue dataset.
- Implemented text preprocessing, tokenization, padding, and truncation using Hugging Face Transformers.
- Fine-tuned T5-small on 4,000 training and 500 validation samples over 6 epochs, and implemented beam-search decoding to generate concise summaries from long conversations.

Movie Review Sentiment Classifier — NLP | Deep Learning

github.com/Karthikvenkat1/Moive_Review_Classifier

Tech Stack: Python, TensorFlow, Keras, LSTM, NLTK

- Developed an LSTM-based sentiment analysis model on the IMDB Movie Reviews dataset.
- Performed text preprocessing, tokenization, sequence padding, and word embedding to classify reviews as positive or negative.
- Tuned model hyperparameters and evaluated performance across training and validation splits.

Price Drop Email Alert System — Automation | Backend

github.com/Karthikvenkat1/PRICE_DROP_ALERT

Tech Stack: Python, FastAPI, SQLAlchemy, BeautifulSoup, APScheduler, Gmail SMTP

- Built a web application to track product prices via web scraping and store price history in a database.
- Automated scheduled price-monitoring jobs using APScheduler and threshold-based comparison logic.
- Implemented automated email notifications through Gmail SMTP when prices dropped below a user-defined threshold.

CERTIFICATIONS & COURSES

- AZ-900: Microsoft Azure Fundamentals — Udemy (Course)
- Introduction to OpenAI GPT Models — Infosys Springboard (Course)
- AWS Solutions Architecture — Forage (Job Simulation)
- GenAI Powered Data Analytics — Forage (Job Simulation)
- Apna College Prime 2.0 – Machine Learning & Data Science (In Progress)

ACHIEVEMENTS

